SAM Rover — User & Operator Guide

Target Audience: Students, Roboticists, Field Operators, Lab Technicians
Web Dashboard Port: `8000` (e.g., `http://<rover-ip>:8000`)

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1. Quick Start Guide (In 60 Seconds)

1. **Power On**: Connect the battery pack or power supply to the Raspberry Pi and L298N motor driver. Wait ~25 seconds for the system to boot.
2. **Open Browser**: On your phone, tablet, or laptop connected to the same WiFi network, navigate to: `http://<rover-ip>:8000`
   (e.g., `http://192.168.0.144:8000` or `http://192.168.0.187:8000`)
3. **Claim Control**: Click the amber `Claim` button in the top navigation bar. The badge will flip to `DRIVER (YOU)` in glowing green.
4. **Drive**: Use **W, A, S, D** or the on-screen joystick to steer the rover!

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2. WiFi & Hotspot Connectivity

The rover uses Linux `NetworkManager` (`nmcli`) for reliable wireless networking in both lab and field environments.

Option A: Connecting Rover to an Existing WiFi (Lab / Home Router)

Connect a screen & keyboard, or SSH over Ethernet/existing connection:

```
# 1. Scan available 2.4GHz & 5GHz WiFi networks
sudo nmcli device wifi list

# 2. Connect to your WiFi network
sudo nmcli device wifi connect "Your_WiFi_SSID" password "Your_WiFi_Password"

# 3. Find the Rover's IP address
ip -br a show wlan0
# or
hostname -I
```

Once connected, the rover will automatically reconnect to this network every time it boots.

Option B: Standalone Autonomous Hotspot Mode (Field Operations Without Router)

When operating outdoors or in an arena without an external WiFi router, configure the Raspberry Pi as a direct WiFi Access Point / Hotspot:

```
# 1. Create and start a WiFi hotspot from the rover
sudo nmcli device wifi hotspot ifname wlan0 ssid "SAM_ROVER_WIFI" password "raspberry"

# 2. Check the hotspot IP address (default is 10.42.0.1)
ip -br a show wlan0
```

To connect:
1. On your smartphone or laptop, open WiFi settings and connect to `SAM_ROVER_WIFI` (password: `raspberry`).
2. Open your web browser and navigate directly to:
`http://10.42.0.1:8000`

To switch back from Hotspot to normal WiFi:

```
# Stop the hotspot and reconnect to your default WiFi network
sudo nmcli connection down Hotspot
sudo nmcli connection up "Your_WiFi_SSID"
```

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3. Web Dashboard & Driving Controls

3.1 Operator Lease System (Multi-Client Arbitration)

To prevent accidental conflicting inputs when multiple users view the same rover:
- **STANDBY (Unclaimed)**: The rover is powered, but motor and LED outputs are hardware-locked. Any student can click `Claim` to become the active driver.
- **DRIVER (YOU)**: You have full control authority. Heartbeats refresh every 250ms.
- **OBSERVER**: Another user is currently driving. You can view the live HD camera feed, telemetry, and sonar radar, but control inputs are safely blocked.
- **RELEASE**: Click the `Release` button when finished. This immediately turns off LEDs, stops PWM, frees all GPIO pins, and allows other users to claim.

3.2 Control Inputs

- **Keyboard (Desktop / Laptop)**:
- `W` / `↑`: Drive Forward
- `S` / `↓`: Drive Reverse
- `A` / `←`: Pivot Left
- `D` / `→`: Pivot Right

- `Spacebar`: Instant Emergency Brake
- `H`: Audio Horn / Buzzer Chirp
- `L`: Toggle Directional LEDs
- **Touch Virtual Joystick (Mobile / Tablet)**:
- Drag the thumbstick on the lower-left for proportional omnidirectional steering.
- **Gamepad / Controller (Xbox / PlayStation / USB HID)**:
- Connect via Bluetooth or USB. Auto-detected with HUD indicator:
  - **Steering**: Left Stick horizontal (`axes[0]`) with 1.4x exponential curve for precision.
  - **Dual Driving Modes**:
  - *RC / Racing Trigger Mode*: Right Trigger (`RT`) accelerates forward, Left Trigger (`LT`) reverses / brakes.
  - *Single-Stick Mode*: Left Stick vertical (`axes[1]`) provides proportional throttle if triggers are idle.
  - **Action Buttons**:
  - `A` **/ Cross**: Audio Horn / Buzzer chirp (hold to sound).
  - `X` **/ Square**: Toggle Directional Headlights / LEDs.
  - `B` **/ Circle**: Instant Emergency Stop (E-Stop).
  - `LB` **/** `RB` **(Bumpers) or D-Pad Up/Down**: Shift speed gears on the fly (`25%` $\leftrightarrow$ `50%` $\leftrightarrow$ `75%` $\leftrightarrow$ `100%`).

3.3 Speed Gears (Authority Selector)

Select gears to match your operational environment:
- `25%` **(Crawl)**: Precision maneuvers, tight doorways, obstacle navigation.
- `50%` **(Standard)**: Indoor laboratory cruising.
- `75%` **(Patrol)**: Fast outdoor driving.
- `100%` **(Sprint)**: Full 60 RPM motor speed and maximum torque.

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4. Safety & Assisted Autonomy Features

SAM is equipped with hardware-level safety layers that intervene automatically:

1. **Active Collision Avoidance**:
2. **Warning Zone (<35 cm)**: Proactively scales down motor speed proportionally to distance.
3. **Critical Zone (≤15 cm)**: Instantly halts forward drive while still permitting steering away or reversing.
4. **IMU Straight-Line Heading-Hold**:
5. Built-in complementary filter gyro assist trims wheel speed asymmetry so the rover drives in a true straight line without drifting.
6. **Teleop Watchdog**:

7. If the browser tab is closed or WiFi packets drop for more than 1.5 seconds while driving, the motors automatically stop.
8. **Idle Auto-Release (180s)**:
9. If no commands are sent for 3 minutes, the rover automatically releases its hardware lease so teammates can use it.

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5. Troubleshooting Matrix

Symptom	Probable Cause	Recommended Action
Cannot open `http://<ip>:8000`	Rover not on network or service down	Check IP via `hostname -I` or router admin. Verify service with `sudo systemctl status sam.service`.
"Hardware unclaimed" warning on click	Rover in STANDBY mode	Click the amber `Claim` button in the top navigation bar to take control.
"Observer mode" message	Another device has claimed the rover	Wait for the driver to click Release, or click `Take Control` to claim the lease.
Motors buzz but do not spin	Battery voltage low or speed gear too low	Check motor battery pack (minimum 7.4V / 2S LiPo). Switch speed gear to 50% or 100%.
Rover drives backward on `W`	Motor wiring polarity inverted	In `config.yaml`, set `invert_left: true` or `invert_right: true` and restart the service.
Camera feed shows test card HUD	USB camera disconnected	Unplug and replug the USB webcam. Verify detection with `ls /dev/video*`.
Ultrasonic sensor reads `--`	Sensor wiring loose or out of range	Check 5V VCC and GND on HC-SR04. Ensure target surface is within 2cm to 300cm range.
LEDs or Horn don't activate	Hardware unclaimed or pin mismatch	Claim the rover first. Verify GPIO pin mapping in `config.yaml`.